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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 18

B.Tech. (EE) (2011 Onwards Elective-II) (Sem.–7)

DIGITAL SIGNAL PROCESSING

Subject Code : BTEE-804C

M.Code : 71938

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

Write briefly :

1. What is an Energy and Power signal?

2. Define recursive and non-recursive system.

3. Determine the fundamental period of the signal.

$$x(n) = \cos\left(\frac{30\pi n}{105}\right)$$

4. Check for following system is stable or unstable.

$$y(n) = x\left(\frac{1}{2n}\right)$$

5. State the convolution property of Fourier Transform.

6. What is ROC in Z-Transform?

7. State the time reversal property of Z-transform.

8. Why the result of circular and linear convolution is not same?

9. What are the various methods to design IIR filters?

10. Write the steps involved in FIR filter design.

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SECTION-B

11. Explain the classification of discrete systems.

12. Find the Z-transform and sketch the ROC :

$$x(n) = a^n \cos \omega_0 nu(n)$$

13. Obtain inverse Z-transform of :

$$X(Z) = \frac{1 - \frac{1}{2} z^{-1}}{1 - \frac{1}{4} z^{-2}} \; |z| > 1/2$$

14. Compute the Fourier Transform of $x(n) = 2^n u(n)$.

15. Determine the length-4 sequence from its DFT :

$$X(K) = [2, 1 - j, 0, 1 + j]$$

SECTION-C

16. The system function of analog filter is as given

$$H_a(S) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

17. Design a FIR low pass filter using Kaiser Window having following specifications :

    Pass-band cut-off frequency = 150 Hz

    Stopband cut-off frequency = 250 Hz

    Passband ripple = 0.1 dB

    Stopband attenuation = 40 dB

    Sampling frequency = 1000 Hz

18. a) State and prove convolution property of DFT.

    b) Determine the length-4 sequence from its DFT

    $X(K) = [2, 1 - j, 0, 1 + j]$

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